

# **Neoadjuvant Chemotherapy with Trastuzumab plus Pertuzumab versus Trastuzumab Alone in Early HER2-Positive Breast Cancer: A Single-Center Retrospective Cohort Study**

*SYNTHETIC DATA FOR TEACHING — NOT A REAL STUDY. Patients, investigators and institution are fictional.*

**This demonstration manuscript was written by AI (Claude Opus 5.5). In a real submission, the text must be written by the authors.**

## **Abstract**

**Background:** Randomized trials showed that adding pertuzumab to neoadjuvant trastuzumab and chemotherapy increases pathologic complete response (pCR) in human epidermal growth factor receptor 2 (HER2)-positive breast cancer, but real-world evidence is limited. We compared dual and single HER2 blockade in routine practice.

**Materials and Methods:** This single-center retrospective cohort study emulated a target trial in patients with clinical stage II–III HER2-positive breast cancer who started neoadjuvant chemotherapy with trastuzumab between January 2012 and December 2023. The primary outcome was pCR (ypT0/is ypN0); isolated tumor cells in lymph nodes were not considered pCR. Because the regimens were used in different years, groups were balanced with propensity-score overlap weighting. Estimates are reported with 95% confidence intervals (CIs).

**Results:** Of 615 patients, 416 received trastuzumab plus pertuzumab and 199 received trastuzumab alone. Unweighted pCR rates were 195/416 (46.9%) and 90/199 (45.2%). After weighting, pCR rates were 57.6% and 40.3% (odds ratio, 2.01 [95% CI, 1.12–3.62];  $P = 0.019$ ). In a post hoc worst-case analysis of patients operated on elsewhere, the odds ratio was 1.40 (95% CI, 0.77–2.53). With a median follow-up of 6.2 years, the hazard ratios were 0.89 (95% CI, 0.47–1.69) for event-free survival and 0.76 (95% CI, 0.34–1.66) for overall survival.

**Conclusion:** In this observational cohort, dual HER2 blockade was associated with a higher pCR rate, consistent with randomized trials, but the estimate depended on adjustment for treatment era and on assumptions about patients without pathology results. The data were insufficient to assess event-free or overall survival.

**Keywords:** Breast neoplasms, Receptor, ErbB-2, Neoadjuvant therapy, Pertuzumab, Trastuzumab, Propensity score

## **Introduction**

Adding pertuzumab to neoadjuvant trastuzumab and chemotherapy increased the rate of pathologic complete response (pCR) in randomized trials of human epidermal growth factor receptor 2 (HER2)-positive breast cancer.<sup>[1,2]</sup> Patients who achieve pCR have longer event-free survival,<sup>[3]</sup> but the long-term benefit of pertuzumab is modest.<sup>[4]</sup> Observational studies have reported higher pCR rates with dual HER2 blockade,<sup>[5]</sup> yet the two regimens were often used in different years, which complicates comparison. We compared neoadjuvant trastuzumab plus pertuzumab with trastuzumab alone in routine practice, using methods designed for limited overlap between treatment groups.

## **Materials and Methods**

### **Study Design and Data Sources**

This single-center retrospective cohort study was designed to emulate a target trial comparing two neoadjuvant treatment strategies.<sup>[6]</sup> Data were extracted from the hospital cancer registry, electronic drug orders, surgical pathology reports, and follow-up records. Before analysis, the extracts were checked for completeness, plausible values, duplicates, and the order of dates; errors identified during these checks were queried with the hospital information office and corrected from source documents. Medical record numbers were replaced with random study identifiers before analysis. The study is reported according to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement,<sup>[7]</sup> and the completed checklist is provided as Supplementary File S2.

## **Patients**

We included patients aged 20 years or older with clinical stage II–III breast cancer (American Joint Committee on Cancer, 8th edition) that was HER2-positive, defined as an immunohistochemistry (IHC) score of 3+ or amplification on in situ hybridization (ISH), who started neoadjuvant chemotherapy with trastuzumab between January 2012 and December 2023. We excluded patients with distant metastasis at diagnosis, surgery before systemic therapy, neoadjuvant trastuzumab emtansine (T-DM1), or surgery at another hospital without an available pathology report. Follow-up data were available until December 31, 2025.

## **Treatment Groups and Outcomes**

The index date was the date of the first neoadjuvant anticancer drug order. Patients were assigned to the dual-blockade group if any pertuzumab order, including the fixed-dose subcutaneous combination of pertuzumab and trastuzumab, occurred between the index date and the day before surgery, and to the single-blockade group otherwise. Time zero for all outcomes was the index date, when eligibility was met and the treatment strategy began. Because pertuzumab could be added after chemotherapy had started, we examined post hoc whether any patient in the single-blockade group had surgery or disease progression before pertuzumab was typically added, which could have introduced immortal time bias.

The primary outcome was pCR, defined as no residual invasive cancer in the breast and axillary lymph nodes (ypT0/is ypN0). In line with international recommendations, isolated tumor cells (ypN0(i+)) and micrometastases in lymph nodes were not considered pCR.<sup>[8,9]</sup> Patients whose disease progressed before surgery were classified as not achieving pCR. The secondary outcome was event-free survival (EFS), measured from the index date to disease progression precluding surgery, locoregional recurrence, distant metastasis, or death from any cause. Overall survival

(OS) was an exploratory outcome. Patients without an event were censored at the later of the last clinic visit and the last drug order, or at the follow-up cutoff, whichever came first.

### **Statistical Analysis**

The statistical analysis plan was finalized before any outcome was examined, and its differences from the original protocol were approved by the institutional review board as a protocol amendment. The propensity score for receiving dual blockade was estimated by logistic regression with 13 baseline covariates: age, menopausal status, Eastern Cooperative Oncology Group performance status, body mass index, baseline left ventricular ejection fraction, clinical T and N categories, hormone receptor status, HER2 IHC score, histologic grade, Ki-67 index, anthracycline-based chemotherapy, and year of treatment start. Because the two regimens were used in largely different periods, we used overlap weighting, which emphasizes patients for whom either treatment was plausible and avoids extreme weights.<sup>[10–12]</sup> Covariate balance was assessed with standardized mean differences (SMDs), with values below 0.1 indicating adequate balance. Missing covariate values were imputed by multiple imputation with 20 data sets that excluded outcome variables, and estimates were combined with Rubin's rules.<sup>[13]</sup>

The pCR rates were compared with weighted logistic regression, reported as odds ratios (ORs), and with weighted risk differences; standard errors accounted for estimation of the propensity score. EFS and OS were estimated with weighted Kaplan-Meier curves and compared with weighted Cox models, reported as hazard ratios (HRs) with robust standard errors; the proportional hazards assumption was assessed with Schoenfeld residuals, and median follow-up was estimated with the reverse Kaplan-Meier method. Prespecified sensitivity analyses restricted the cohort to patients who started treatment in 2016–2019, used one-to-one propensity-score matching, used multivariable regression, defined pCR as ypT0 ypN0, and included only patients

with complete covariate data. An exploratory analysis tested whether the association with pCR differed by hormone receptor status. The E-value was calculated to quantify how strong unmeasured confounding would need to be to explain away the primary result.<sup>[14]</sup> Two-sided *P* values below 0.05 were considered statistically significant. Analyses were performed with R version 4.5.1 using the WeightIt, MatchIt, cobalt, mice, survival, and EValue packages.

In response to peer review, we performed the following post hoc analyses, which were not part of the prespecified plan: the distribution of propensity scores by group and the share of the overlap weights contributed by each treatment period; a propensity-score model with a natural spline for year of treatment start; EFS restricted to patients who started treatment in 2012–2019, before adjuvant T-DM1 was used at the study hospital, instead of stratification by post-neoadjuvant T-DM1, which would condition on a consequence of the response to neoadjuvant therapy; pCR after exclusion of patients whose disease progressed before surgery; best-case and worst-case analyses that included the patients operated on at another hospital, assigned to groups by pertuzumab use within 180 days of the index date and assumed either all to achieve or all not to achieve pCR; and EFS and OS with follow-up truncated at five years, together with the five-year restricted mean survival time (RMST) difference. In response to the second review, we also compared the weighted distribution of treatment periods under the prespecified and spline propensity-score models.

## **Ethics**

The study was approved by the institutional review board of the participating hospital, which waived the requirement for informed consent because of the retrospective design and the use of de-identified data. The study was conducted in accordance with the Declaration of Helsinki (1975, revised in 2013).

## Results

### Patients

Of 703 registry records, three duplicate records were removed, leaving 700 patients (Figure 1).

After exclusion of patients with clinical stage 0–I disease ( $n = 27$ ), distant metastasis at diagnosis ( $n = 14$ ), HER2-negative disease ( $n = 11$ ), no systemic therapy at the study hospital ( $n = 17$ ), and surgery at another hospital ( $n = 16$ ), 615 patients were included: 416 received trastuzumab plus pertuzumab and 199 received trastuzumab alone.

Baseline characteristics were similar between the groups except for the year of treatment start (SMD, 2.80) and the use of anthracycline-based chemotherapy (SMD, 0.55) (Table 1). Most patients who received trastuzumab alone started treatment in 2012–2015, whereas most patients who received dual blockade started treatment in 2020–2023. The propensity scores of the two groups overlapped only partly (median, 0.99 with dual blockade vs. 0.05 with trastuzumab alone; Supplementary Figure S2): 64.9% and 71.9% of the patients, respectively, had scores within the range of the other group, and patients who started treatment in 2016–2019 contributed 67.6% of the overlap weights (post hoc). After overlap weighting, every covariate in the propensity-score model was exactly balanced (maximum absolute SMD, 0.000), and the effective sample sizes were 86 and 66 patients, respectively. Because year of treatment start was modeled as a continuous variable, weighting balanced its mean but not the distribution of treatment periods (SMD, 0.67; Table 1): 13.7% of the weighted dual-blockade group and 32.8% of the weighted single-blockade group started treatment in 2012–2015 (post hoc). The median follow-up was 6.2 years overall, 5.1 years for dual blockade, and 10.7 years for trastuzumab alone.

### **Pathologic Complete Response**

In the unweighted data, pCR was achieved in 195/416 (46.9%) patients who received dual blockade and in 90/199 (45.2%) patients who received trastuzumab alone (Table 2). Within treatment periods, however, crude pCR rates were higher with dual blockade (2012–2015: 3/6 (50.0%) vs. 73/156 (46.8%); 2016–2019: 90/171 (52.6%) vs. 17/38 (44.7%)), and the overall pCR rate in 2020–2023, when only five patients received trastuzumab alone, was 102/244 (41.8%). After overlap weighting, the pCR rate was 57.6% with dual blockade and 40.3% with trastuzumab alone, a difference of 17.3 percentage points (95% CI, 3.1–31.5); the OR was 2.01 (95% CI, 1.12–3.62) ( $P = 0.019$ ).

### **Event-Free and Overall Survival**

EFS events occurred in 62 patients who received dual blockade and in 68 patients who received trastuzumab alone. The weighted five-year EFS rates were 88.6% and 81.2%, respectively (HR, 0.89 [95% CI, 0.47–1.69]) (Table 3, Figure 2). Deaths occurred in 34 and 53 patients, and the weighted five-year OS rates were 89.6% and 86.9% (HR, 0.76 [95% CI, 0.34–1.66]) (Figure 3). The five-year RMST difference was 0.17 years (95% CI, –0.39 to 0.73) for EFS and 0.08 years (95% CI, –0.30 to 0.47) for OS. The confidence interval for OS was wide and compatible with both a clinically important benefit and harm. There was no evidence against proportional hazards for EFS ( $P = 0.394$ ) or OS ( $P = 0.462$ ). Among patients who did not achieve pCR, adjuvant T-DM1 was given to 59.7% in the dual-blockade group and 4.6% in the single-blockade group.

### **Sensitivity and Subgroup Analyses**

The direction of the association with pCR was consistent across sensitivity analyses (Table 2): the OR was 1.97 (95% CI, 1.10–3.53) with multivariable regression and 2.38 (95% CI, 1.27–4.46) when pCR was defined as ypT0 ypN0. In the cohort restricted to 2016–2019 (171 and 38

patients) and in the complete-case analysis, the ORs were 2.15 (95% CI, 0.92–5.01) and 1.91 (95% CI, 0.99–3.68), respectively, with confidence intervals that included the null value. Propensity-score matching retained 44 pairs and yielded a smaller estimate (OR, 1.33 [95% CI, 0.52–3.43]). The association did not differ by hormone receptor status (receptor-negative OR, 2.63 [95% CI, 0.79–8.71]; receptor-positive OR, 2.03 [95% CI, 0.94–4.37];  $P = 0.552$  for interaction). The E-value for the primary OR was 2.19, and that for its lower confidence limit was 1.31.

### **Post Hoc Analyses**

In post hoc analyses (Supplementary Table S4), the OR for pCR was 2.00 (95% CI, 1.05–3.79) when the year of treatment start was modeled with a natural spline, under which the SMD for treatment period after weighting was 0.17, and 2.01 (95% CI, 1.12–3.62) after exclusion of the patients whose disease progressed before surgery (four and five patients, respectively). When the six dual-blockade and ten single-blockade patients who were operated on at another hospital were included, the OR was 2.19 (95% CI, 1.27–3.77) under the assumption that none of them achieved pCR and 1.40 (95% CI, 0.77–2.53) under the assumption that all did. Among patients who started treatment in 2012–2019 (177 and 194 patients), none of those without pCR received adjuvant T-DM1, and the HR for EFS was 0.81 (95% CI, 0.39–1.70). With follow-up truncated at five years, the HRs were 0.59 (95% CI, 0.26–1.36) for EFS and 0.77 (95% CI, 0.30–1.99) for OS.

Pertuzumab was started on the index date in 300 patients and later in 116 (median, day 84), whereas the earliest surgery or disease progression in the single-blockade group occurred on day 97, so no patient could have been classified as receiving single blockade because of an event before pertuzumab would have been added.

## Discussion

In this single-center cohort, neoadjuvant trastuzumab plus pertuzumab was associated with a higher pCR rate than trastuzumab alone after measured characteristics were balanced with overlap weighting. The size of the difference was similar to that observed in randomized trials, but the evidence for EFS and OS was inconclusive.

In NeoSphere, adding pertuzumab to trastuzumab and docetaxel increased the breast pCR rate from 29.0% to 45.8%,<sup>[1]</sup> and the Asian PEONY trial reported a similar increase in total pCR,<sup>[2]</sup> with five-year EFS of 84.8% versus 73.7% in its descriptive final analysis.<sup>[15]</sup> In TRYPHAENA, dual blockade combined with anthracycline-containing or carboplatin-based chemotherapy produced breast pCR rates of 57.3%–66.2%,<sup>[16]</sup> similar to our weighted rate, although that trial defined pCR in the breast only. Real-world studies from Turkey, Italy, China, and Spain have also reported higher pCR rates with dual blockade,<sup>[17–20]</sup> and a meta-analysis of 18 real-world studies estimated a pooled OR of 1.81 (95% CI, 1.56–2.09).<sup>[5]</sup> Our estimate is consistent with these findings.

Our results also show how strongly treatment era can influence real-world comparisons. The two regimens were used in largely different years, and the crude pCR rates were almost identical because the overall pCR rate was lower in recent years, when nearly all patients received dual blockade. Within each period, however, pCR was more frequent with dual blockade. We could not determine why the pCR rate was lower in recent years; possible explanations include changes in patient selection, chemotherapy backbones, and pathologic assessment, such as more thorough examination of lymph nodes that identifies isolated tumor cells as residual disease. Overlap weighting addresses this problem by focusing the comparison on patients who were treated when either regimen was plausible, mainly those treated in 2016–2019, and avoids the extreme weights that inverse probability weighting would have produced.<sup>[11]</sup> Post hoc analyses supported this

approach: the propensity scores overlapped only partly, the comparison drew mainly on patients treated in 2016–2019, and modeling the year of treatment start more flexibly did not change the estimate. The prespecified model balanced the mean year of treatment start but not the distribution of treatment periods (SMD, 0.67); the more flexible model reduced this imbalance (SMD, 0.17) and gave a similar estimate. The estimate should therefore be interpreted as an association in patients for whom both regimens were realistic options.

The EFS and OS results were inconclusive. Follow-up was shorter in the dual-blockade group, events were few, and post-neoadjuvant treatment changed over time. After the KATHERINE trial showed that adjuvant T-DM1 improves invasive disease-free survival and OS in patients with residual disease,<sup>[21,22]</sup> T-DM1 was used mainly in the dual-blockade era in our cohort. In APHINITY, adjuvant pertuzumab did not improve OS at a median follow-up of 8.4 years,<sup>[4]</sup> and pCR has not been validated as a trial-level surrogate for EFS or OS.<sup>[3]</sup> In a post hoc analysis restricted to patients who started treatment before adjuvant T-DM1 was used, the HR for EFS was similar to that in the full cohort, which argues against post-neoadjuvant T-DM1 as the main explanation of our EFS estimate; we did not stratify by T-DM1 because its use depends on the response to neoadjuvant therapy. The confidence intervals for EFS and OS were wide, and the OS results in particular should be regarded as exploratory. A higher pCR rate should therefore not be assumed to translate into a proportional survival benefit.

We applied the international definition of pCR, under which residual isolated tumor cells are considered residual disease.<sup>[8,9]</sup> Low-volume residual nodal disease after neoadjuvant chemotherapy, including isolated tumor cells, has been associated with worse disease-free and overall survival than node-negative disease,<sup>[23]</sup> which supports this stricter definition.

The strengths of this study include an analysis plan that was finalized before outcomes were examined, validation and correction of the source data before analysis, a design that addressed

the limited overlap between treatment eras, and several sensitivity analyses. This study also has limitations. First, as a single-center retrospective study, the results reflect one institution's patient population, treatment practices, and pathology procedures and may not apply to other settings, and residual confounding by unmeasured factors associated with treatment era cannot be excluded, particularly because the distribution of treatment periods remained imbalanced after weighting; an unmeasured confounder associated with both treatment and pCR by a risk ratio of approximately 2.19, beyond the measured covariates, could explain the observed association. Second, the effective sample size after weighting was small, so the estimates were imprecise, and the matched analysis, which retained only 44 pairs, yielded a smaller estimate. Third, patients operated on at another hospital had no pathology results; in a post hoc worst-case analysis that assumed all of them achieved pCR, the association was attenuated and its confidence interval included the null value. Fourth, EFS did not include contralateral breast cancer or second primary cancers because these events were not recorded, and loss to follow-up may have differed between treatment eras. Fifth, data on other active malignancies and on participation in clinical trials were unavailable. Finally, missing covariate data were handled by multiple imputation, which assumes that the data were missing at random, and the post hoc analyses were not prespecified.

In conclusion, in this observational cohort, adding pertuzumab to neoadjuvant trastuzumab and chemotherapy was associated with a higher pCR rate, consistent with randomized evidence. Interpreting this association as an effect of the regimen assumes that no unmeasured factor influenced both treatment choice and pCR and that every patient could have received either regimen (positivity), an assumption that was only partly met because the regimens were used in different years. The result depended on adjustment for treatment era and on assumptions about patients without pathology results, and the data were insufficient to determine whether dual blockade is associated with longer EFS or OS.

## **Data Availability Statement**

The datasets generated and/or analyzed during the current study are not publicly available due to patient privacy regulations but are available from the corresponding author on reasonable request. In this teaching demonstration, all data are synthetic.

## **Use of Generative AI**

This demonstration manuscript, including all of its text, was written by an AI model (Claude Opus 5.5, Anthropic) at the request of the authors for teaching purposes; in a real submission, the text must be written by the authors, as required by the journal's policy on generative AI. Claude Opus 5.5 was also used, under the authors' direction, to write and test the statistical analysis code, check the data, produce the tables and figures, and format the manuscript. OpenEvidence was used to search the literature, and every reference was verified against Crossref and PubMed. The analysis plan was specified by the authors before outcomes were examined and was approved by the institutional review board. The authors are responsible for reviewing all code, results, and text and take full responsibility for the content. The analysis code can be provided as supplementary material.

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**Table 1. Baseline characteristics of the study cohort, by neoadjuvant anti-HER2 regimen**

| Characteristic                                   | Trastuzumab +<br>pertuzumab (n =<br>416) | Trastuzumab<br>alone (n = 199) | SMD,<br>unweighted <sup>†</sup> | SMD,<br>weighted <sup>‡</sup> |
|--------------------------------------------------|------------------------------------------|--------------------------------|---------------------------------|-------------------------------|
| Age at diagnosis, years, median (IQR)            | 50.7 (43.8–59.0)                         | 51.5 (42.6–57.7)               | 0.05                            | 0.00                          |
| Postmenopausal                                   | 232 (55.8)                               | 116 (58.3)                     | 0.05                            | 0.00                          |
| ECOG performance status $\geq 1$                 | 107 (25.7)                               | 58 (29.1)                      | 0.08                            | 0.00                          |
| BMI, kg/m <sup>2</sup> , median (IQR)            | 24.1 (21.5–26.8)                         | 24.0 (21.2–26.4)               | 0.08                            | 0.00                          |
| Baseline LVEF, %, median (IQR)                   | 65.0 (61.0–68.0)                         | 65.0 (62.0–68.0)               | 0.08                            | 0.00                          |
| Clinical T category                              |                                          |                                | 0.09                            | 0.00                          |
| T1                                               | 32 (7.7)                                 | 17 (8.5)                       |                                 |                               |
| T2                                               | 237 (57.0)                               | 111 (55.8)                     |                                 |                               |
| T3                                               | 97 (23.3)                                | 42 (21.1)                      |                                 |                               |
| T4                                               | 50 (12.0)                                | 29 (14.6)                      |                                 |                               |
| Clinical N category                              |                                          |                                | 0.00                            | 0.00                          |
| N0                                               | 122 (29.3)                               | 58 (29.1)                      |                                 |                               |
| N1                                               | 196 (47.1)                               | 94 (47.2)                      |                                 |                               |
| N2                                               | 56 (13.5)                                | 27 (13.6)                      |                                 |                               |
| N3                                               | 42 (10.1)                                | 20 (10.1)                      |                                 |                               |
| Hormone receptor positive (ER or PR $\geq 1\%$ ) | 240 (57.7)                               | 116 (58.3)                     | 0.01                            | 0.00                          |
| HER2 IHC 3+ (vs IHC 2+, ISH amplified)           | 320 (76.9)                               | 155 (77.9)                     | 0.02                            | 0.00                          |
| Histologic grade 3*                              | 201 (51.9)                               | 88 (48.6)                      | 0.07                            | 0.00                          |
| Ki-67, %, median (IQR)*                          | 43.5 (31.2–56.0)                         | 43.5 (30.0–56.8)               | 0.05                            | 0.00                          |
| Anthracycline-based chemotherapy                 | 116 (27.9)                               | 107 (53.8)                     | 0.55                            | 0.00                          |
| Year of neoadjuvant therapy start                |                                          |                                | 2.80                            | 0.67                          |
| 2012–2015                                        | 6 (1.4)                                  | 156 (78.4)                     |                                 |                               |
| 2016–2019                                        | 171 (41.1)                               | 38 (19.1)                      |                                 |                               |
| 2020–2023                                        | 239 (57.5)                               | 5 (2.5)                        |                                 |                               |

Values are n (%) unless otherwise indicated. Abbreviations: BMI, body mass index; ECOG,

Eastern Cooperative Oncology Group; ER, estrogen receptor; HER2, human epidermal growth

factor receptor 2; IHC, immunohistochemistry; IQR, interquartile range; ISH, in situ

hybridization; LVEF, left ventricular ejection fraction; PR, progesterone receptor; SMD, standardized mean difference.

\*Missing: histologic grade, 47; Ki-67, 71. Percentages exclude missing values.

†Before weighting, in patients with available data.

‡After overlap weighting, averaged over the 20 imputed data sets. Year of treatment start entered the propensity-score model as a continuous variable, so its mean was exactly balanced but the distribution of treatment periods was not.

**Table 2. Pathologic complete response: primary and sensitivity analyses**

| Analysis                           | Patients,<br>n<br>(T+P/T) | pCR, T+P   | pCR, T    | Risk difference,<br>percentage<br>points (95% CI) | Odds ratio<br>(95% CI) | P     |
|------------------------------------|---------------------------|------------|-----------|---------------------------------------------------|------------------------|-------|
| Unweighted                         | 416/199                   | 195 (46.9) | 90 (45.2) |                                                   |                        |       |
| Overlap weighting (primary)        | 416/199                   | 57.6%*     | 40.3%*    | 17.3 (3.1–31.5)                                   | 2.01 (1.12–3.62)       | 0.019 |
| S1: Treatment start 2016–2019 only | 171/38                    | 63.1%*     | 44.3%*    | 18.8 (–1.7 to 39.3)                               | 2.15 (0.92–5.01)       | 0.076 |
| S2: 1:1 propensity-score matching  | 44/44                     | 53.9%*     | 46.8%*    | 7.1 (–16.4 to 30.6)                               | 1.33 (0.52–3.43)       | 0.552 |
| S3: Multivariable regression       | 416/199                   |            |           |                                                   | 1.97 (1.10–3.53)       | 0.024 |
| S4: pCR defined as ypT0 ypN0       | 416/199                   | 40.4%*     | 22.1%*    | 18.2 (5.6–30.9)                                   | 2.38 (1.27–4.46)       | 0.007 |
| S6: Complete cases                 | 333/164                   | 56.2%*     | 40.3%*    | 16.0 (0.0–32.0)                                   | 1.91 (0.99–3.68)       | 0.054 |

Unweighted row: n (%) of patients with pCR. Abbreviations: CI, confidence interval; pCR, pathologic complete response (ypT0/is ypN0; isolated tumor cells in lymph nodes were not considered pCR); T, trastuzumab; T+P, trastuzumab plus pertuzumab.

\*Weighted (overlap weighting) or matched proportions, averaged over 20 imputations; no absolute counts correspond to weighted proportions.

S3 adjusts for the propensity-score covariates by regression and has no corresponding weighted proportions or risk difference.

**Table 3. Event-free and overall survival**

| Analysis                           | Events, n<br>(T+P/T) | Hazard ratio<br>(95% CI) | <i>P</i> | 5-year<br>rate, T+P | 5-year<br>rate, T |
|------------------------------------|----------------------|--------------------------|----------|---------------------|-------------------|
| Event-free survival†               |                      |                          |          |                     |                   |
| Overlap weighting (primary)        | 62/68                | 0.89 (0.47–1.69)         | 0.718    | 88.6%*              | 81.2%*            |
| S1: Treatment start 2016–2019 only | 37/9                 | 0.74 (0.29–1.90)         | 0.536    | 89.7%*              | 75.7%*            |
| S2: 1:1 propensity-score matching  | 9.3‡/11.2‡           | 0.85 (0.34–2.13)         | 0.725    |                     |                   |
| S3: Multivariable regression       | 62/68                | 0.94 (0.53–1.70)         | 0.847    |                     |                   |
| S6: Complete cases                 | 47/56                | 1.00 (0.48–2.07)         | 0.998    | 91.0%*              | 85.1%*            |
| Overall survival                   |                      |                          |          |                     |                   |
| Overlap weighting (primary)        | 34/53                | 0.76 (0.34–1.66)         | 0.488    | 89.6%*              | 86.9%*            |
| S1: Treatment start 2016–2019 only | 21/8                 | 0.56 (0.19–1.61)         | 0.282    | 90.1%*              | 82.7%*            |
| S2: 1:1 propensity-score matching  | 7.3‡/10.8‡           | 0.75 (0.28–2.04)         | 0.571    |                     |                   |
| S3: Multivariable regression       | 34/53                | 0.68 (0.32–1.45)         | 0.321    |                     |                   |
| S6: Complete cases                 | 26/43                | 0.94 (0.38–2.33)         | 0.889    | 91.6%*              | 87.6%*            |

Abbreviations: CI, confidence interval; T, trastuzumab; T+P, trastuzumab plus pertuzumab.

\*Weighted Kaplan-Meier estimates, averaged over 20 imputations.

†Events: progression precluding surgery, locoregional recurrence, distant metastasis, or death from any cause.

‡Matched analysis: mean number of events across imputations.

## Figure Legends

**Figure 1.** Flow diagram of patient selection. HER2, human epidermal growth factor receptor 2.

**Figure 2.** Overlap-weighted Kaplan-Meier curves of event-free survival by neoadjuvant regimen, averaged over 20 imputations, with the numbers of patients at risk. CI, confidence interval; HR, hazard ratio.

**Figure 3.** Overlap-weighted Kaplan-Meier curves of overall survival by neoadjuvant regimen, averaged over 20 imputations, with the numbers of patients at risk. CI, confidence interval; HR, hazard ratio.